

A range of practical experiences is a vital part of a learner's development as part of this course.

Learners should develop and practise a wide range of practical skills throughout the course as preparation for the Practical Endorsement, as well as for the written examinations.

The experiments and skills required for the Practical Endorsement will allow learners to develop and practise their practical skills, preparing learners for the written examinations.

Please refer to Section 5g (the Practical Endorsement) of this specification to see the list of practical experiences all learners should cover during the course. Further advice and guidance on the Practical Endorsement can be found in the Practical Skills Handbook support booklet.

1.2.1 Practical skills

Learning outcomes	Additional guidance
<i>Practical work carried out throughout the course will enable learners to develop the following skills:</i>	
Independent thinking	
(a) apply investigative approaches and methods to practical work	Including how to solve problems in a practical context. HSW3
Use and application of scientific methods and practices	
(b) safely and correctly use a range of practical equipment and materials	See Section 5g. Including identification of potential hazards. Learners should understand how to minimise the risks involved. HSW4
(c) follow written instructions	HSW4
(d) make and record observations/measurements	HSW8
(e) keep appropriate records of experimental activities	See Section 5g.
(f) present information and data in a scientific way	HSW8
(g) use appropriate software and tools to process data, carry out research and report findings	M3.1 HSW3
Research and referencing	
(h) use online and offline research skills including websites, textbooks and other printed scientific sources of information	

- (i) correctly cite sources of information

The Practical Skills Handbook provides guidance on appropriate methods for citing information.

Instruments and equipment

- (j) use a wide range of experimental and practical instruments, equipment and techniques appropriate to the knowledge and understanding included in the specification.

See Section 5g.
HSW4